

```
/*  
  
#include <iostream>  
  
using namespace std;  
  
int main() //1  
{  
    string name;  
    int unit;  
    float charge=0,surcharge;  
  
    cout<<"Enter name of the consumer : ";  
    cin>>name;  
    cout<<"Enter Units Consumed : ";  
    cin>>unit;  
  
    if(unit>300)  
    {  
        charge=(unit-300)*0.90;  
        unit=300;  
    }  
  
    if(unit>100)  
    {  
        charge+=(unit-100)*0.80;  
        unit=100;  
    }  
  
    if(unit>0)  
    charge+=unit*0.60;  
  
    if(charge>300)  
    surcharge=charge*15/100;  
  
    if(surcharge<50)  
    surcharge=50;  
  
    charge+=surcharge;  
  
    cout<<"Name : "<<name<<endl<<"Charges : "<<charge<<endl;  
    return 0;  
}  
  
*/
```

```
/*  
  
#include <iostream>  
  
using namespace std;
```

```
int main() //2
{
    string a;
    int j=0;

    cout<<"Input : ";
    getline(cin,a);

    char *b = (char *)malloc(a.size()*sizeof(char));

    for(int i=0;i<a.size();i++)
    {
        if(a[i]!='t')
        {
            b[j]=a[i];
            j++;
        }
    }
    b[j]='\0';

    cout<<"Output : "<<b<<endl;

    free(b);

    return 0;
}

*/
```

```
/*
#include <iostream>

using namespace std;

int main() //3
{
    char a;
    int x,z,k=0;
    string s;

    cout<<"Input String : ";
    getline(cin,s);

    char *t = (char *)malloc(s.size()*sizeof(char));

    for(int i=0;i<s.size();i++)
    {
        if(s[i]=='#')
            continue;
    }
}
```

```

        z=0;
        a=s[i];
        for(int j=i+1;j<s.size();j++)
        {
            if(s[j]=='#')
                continue;
            if(a==s[j])
            {
                z++;
                s[j]='#';
            }
        }
        s[i]='#';
        if(z==0)
        {
            t[k]=a;
            k++;
        }
    }

    for(int i=0;i<k-1;i++)
    {
        for(int j=i+1;j<k;j++)
        {
            if(t[i]>t[j])
            {
                a=t[i];
                t[i]=t[j];
                t[j]=a;
            }
        }
    }

    for(int i=0;i<k;i++)
        cout<<t[i]<<' ';

    cout<<endl;

    free(t);

    return 0;
}

*/

```

```

/*

```

```

#include <iostream>

```

```
#include <cstdlib>

using namespace std;

int main()
{
    int n,*a,*s,z=0;

    cout<<"Size : ";
    cin>>n;

    a = (int *)malloc(n*sizeof(int));
    s = (int *)malloc((n-2)*sizeof(int));

    cout<<"Elements : ";
    for(int i=0;i<n;i++)
        cin>>a[i];

    t=s[0];
    for(int i=1;i<n-2;i++)
    {
        if(t<s[i])
            t=s[i];
    }
    cout<<t<<endl;

    free(a);
    free(s);

    return 0;
}

*/
```

```
/*
```

```
#include <iostream>

using namespace std;
```

```
int main()
{
    int c,n,*a,t,s,l=0,z;

    cout<<"Size : ";
    cin>>n;
```

```
//4
```

```

a=(int *)malloc(n*sizeof(int));

cout<<"Elements : ";
for(int i=0;i<n;i++)
cin>>a[i];

for(int i=0;i<n;i++)
{
    s=0,c=1,t=n-i,z=i;
    while(t>=c)
    {
        for(int j=0;j<c;j++,z++)
            s+=a[z];
        t-=c;
        c++;
    }
    if(l<s)
        l=s;
}
cout<<"Output : "<<l<<endl;

free(a);
return 0;
}

*/

```

```

/*

#include <iostream>

using namespace std;

int main()
{
    int c,n,*a,t,s,l=0,z,y;

    cout<<"Size : ";
    cin>>n;

    a=(int *)malloc(n*sizeof(int));

    cout<<"Elements : ";
    for(int i=0;i<n;i++)
        cin>>a[i];

```

```

for(int i=0;i<n;i++)
{
    s=0,c=1,t=n-i,z=i;
    cout<<"t : "<<t<<endl;
    while(t>=c)
    {
        y=0;
        cout<<"c : "<<c<<endl;
        for(int j=0;j<c;j++,z++)
        {
            cout<<"y : ";
            y+=a[z];
            cout<<y<<endl;
        }
        s+=y;
        cout<<"s : "<<s<<endl;
        t-=c;
        c++;
        cout<<"t : "<<t<<endl;
    }
    cout<<"c : "<<c<<endl;
    if(l<s)
    l=s;
    cout<<s<<" "<<l<<" "<<endl;
}

cout<<"Output : "<<l<<endl;

free(a);
return 0;
}

*/

```

```

/*

```

```

#include <iostream>

```

```

using namespace std;                                     //5

```

```

class car                                                //data abstraction
{
    private :                                           // Class and Objects
        int speed;

    public :

```

```
        void set(int s)
        {
            speed=s;
        }

        int get()
        {
            return speed;
        }

};

int main()
{
    int s;
    car a;

    cout<<"Speed : ";
    cin>>s;

    a.set(s);
    cout<<a.get()<<endl;

    return 0;
}

*/

/*
#include <iostream>

using namespace std;

class hotel //6
{
    private :
        int rno,nod;
        float t;
        string n;

        float calc(float t,int nod)
        {
            float z=t*nod;

            if(z>10000)
```

```
        return 1.05*z;

        else
            return z;
    }

public :
    void checkin()
    {
        cout<<"Room No. ";
        cin>>rno;

        cout<<"Name : ";
        cin>>n;

        cout<<"Tarrif : ";
        cin>>t;

        cout<<"No of Days : ";
        cin>>nod;
    }

    void checkout()
    {
        cout<<rno<<endl<<n<<endl<<t<<endl<<nod<<endl;

        cout<<calc(t,nod)<<endl;
    }

};

int main()
{
    hotel a;

    a.checkin();

    a.checkout();

    return 0;
}

*/

/*
```



```
#include <iostream>

using namespace std;

class bank //7
{
    public:
        string n;
        int acc,bal;
        char t;

        void assign()
        {
            cout<<"Name : ";
            cin>>n;

            cout<<"Acc no. ";
            cin>>acc;

            cout<<"Account Type : ";
            cin>>t;

            cout<<"Balance : ";
            cin>>bal;

        }

        int deposit()
        {
            int z;

            cout<<"Amount to Deposit : ";
            cin>>z;

            bal+=z;

            return bal;

        }

        int withdraw()
        {
            int y;

            cout<<"Current Balance : "<<bal<<endl;

            cout<<"Amount to Withdraw : ";
            cin>>y;

            bal-=y;

            return bal;

        }
}
```

```
        void display()
        {
            cout<<n<<endl<<bal<<endl;
        }

};

int main()
{
    bank b;

    b.assign();

    b.deposit();

    b.withdraw();

    b.display();

    return 0;
}

*/

/*
#include <iostream>

using namespace std;

class student //8
{
    public :
        int s[5];

        void input()
        {
            for(int i=0;i<5;i++)
                cin>>s[i];
        }

        int calc()
        {
            int sum=0;

            for(int i=0;i<5;i++)
```

```
                sum+=s[i];
                return sum;
            }

};

int main()
{
    student a;
    int n,c=0,anna=136;

    cout<<"No. of students in class : ";
    cin>>n;

    for(int i=0;i<n;i++)
    {
        a.input();

        if(anna<a.calc())
            c++;
    }

    cout<<c<<endl;

    return 0;
}

*/

/*
#include <iostream>
#include <cmath>

using namespace std;

class ar //9
{
    public :
        int a,b,c;

        int area(int a)
        {
            return a*a;
        }

        int area(int a,int b)
        {
            return a*b;
        }
    };
};
```

```

    }

    int area(int a,int b,int c)
    {
        int s=a+b+c,z;
        z=s*(s-a)*(s-b)*(s-c);
        return sqrt(z);
    }
};

int main()
{
    int a,b,c,d,e,f;

    cin>>a>>b>>c>>d>>e>>f;

    ar square,rectangle,triangle;

    cout<<square.area(a)<<endl;
    cout<<rectangle.area(b,c)<<endl;
    cout<<triangle.area(d,e,f)<<endl;

    return 0;
}

*/

/*
#include <iostream>
#include <string>

using namespace std;

class invoice //10
{
    private :

        string no,des;
        int q;
        double pq;

    public :

        invoice()
        {
            q=0,pq=0;
            no="M000";
        }
    };

```

```
        des="Hardware Part";
    }

    double amount() {return q*pq;}

    string get_no()          {return no;}
    void set_no(string b)    {no=b;}

    string get_des()         {return des;}
    void set_des(string b)   {des=b;}

    int get_q()              {return q;}
    void set_q(int b)         {if(b>0) q=b;}

    double get_pq()          {return pq;}
    void set_pq(double b)    {if(b>0) pq=b;}

    void input()
    {
        int i;
        double j;
        string k;

        cout<<"no : ";
        getline(cin,k);
        set_no(k);

        cout<<"des : ";
        getline(cin,k);
        set_des(k);

        cout<<"q : ";
        cin>>i;
        set_q(i);

        cout<<"pq : ";
        cin>>j;
        set_pq(j);
    }

};

int main()
{
    invoice y;
```

```

        cout<<y.get_no()<<endl<<y.get_des()<<endl;
        cout<<y.get_q()<<endl<<y.get_pq()<<endl;

        cout<<"Amount : "<<y.amount()<<endl<<endl;

        y.input();

        cout<<y.get_no()<<endl<<y.get_des()<<endl;
        cout<<y.get_q()<<endl<<y.get_pq()<<endl;

        cout<<"Amount : "<<y.amount()<<endl;

        return 0;
    }
    */

```

```

/*
#include<iostream>

using namespace std;

class tollbooth                                //11
{
    unsigned int c;
    double m;
public:
    tollbooth()
    {
        m=0;
        c=0;
    }

    void paying_cars()
    {
        c++;
        m+=0.5;
    }

    void non_paying_cars()
    {
        c++;
    }

    void display()
    {

```

```

        cout<<"Cars : "<<c<<endl;
        cout<<"Cash : "<<m<<endl;
    }

};

```

```

int main()
{
    tollbooth a;
    char n;
    do
    {
        printf("Choice : ");
        cin>>n;
        switch(n)
        {
            case '1':
                a.paying_cars();
                break;
            case '2':
                a.non_paying_cars();
                break;
            case '3':
                a.display();
                break;
            case 27:
                break;
            default:
                printf("Invalid!\n");
        }
    }while(n!=27);

    return 0;
}

*/

```

```

/*

```

```

#include <iostream>

```

```

using namespace std;

```

```

class time
{

```

```

//12

```

```
private :  
    int h,m,s;  
public :  
    time()  
    {  
        h=m=s=0;  
    }  
    time(int a,int b,int c)  
    {  
        h=a,m=b,s=c;  
    }  
    time add(time p,time q)  
    {  
        time r;  
  
        r.h=p.h+q.h;  
        r.m=p.m+q.m;  
        r.s=p.s+q.s;  
  
        if(r.s>=60)  
        {  
            r.m++;  
            r.s-=60;  
        }  
        if(r.m>=60)  
        {  
            r.h++;  
            r.m-=60;  
        }  
  
        return r;  
    }  
    void display()  
    {  
        if(h/10==0)  
            cout<<"0"<<h;  
        else  
            cout<<h;  
        cout<<":";  
  
        if(m/10==0)  
            cout<<"0"<<m;  
        else  
            cout<<m;  
        cout<<":";  
  
        if(s/10==0)  
            cout<<"0"<<s;
```



```
        else
        cout<<s;
        cout<<endl;
    }

};

int main()
{
    class time x(7,10,45),y(10,25,40),z;
    z=z.add(x,y);
    z.display();
    return 0;
}

*/

/*
#include <iostream>
using namespace std;
class acc //13
{
    private :
        double bal,i;
    public :
        void set_bal(int x)
        {
            bal=x;
        }
        void set_i(int x)
        {
            i=x;
        }
        double month()
        {
            return (bal*i)/1200.0;
        }
}
```

```

        double total()
        {
            bal+=month();
            return bal;
        }

};

int main()
{
    acc a,b;
    double n,p,q;

    cin>>p>>q;                //2000 3000

    a.set_bal(p);
    b.set_bal(q);

    cin>>n;                    //4

    a.set_i(n);
    b.set_i(n);

    cout<<a.total()<<endl;
    cout<<b.total()<<endl;

    cin>>n;                    //5

    a.set_i(n);
    b.set_i(n);

    cout<<a.total()<<endl;
    cout<<b.total()<<endl;

    return 0;
}

*/

/*
#include <iostream>

using namespace std;

class complex                //14
{
    double r,i;

```

```
public :

    complex()
    {
        r=0,i=0;
    }

    complex(double x,double y)
    {
        r=x,i=y;
    }

    complex operator+(complex x)
    {
        r+=x.r;
        i+=x.i;
        return complex(r,i);
    }

    complex operator-(complex x)
    {
        r-=x.r;
        i-=x.i;
        return complex(r,i);
    }

    bool operator==(complex x)
    {
        if(r==x.r && i==x.i)
            return true;
        else
            return false;
    }

    void display()
    {
        cout<<r<<" + i" << endl;
    }

    void input()
    {
        cin>>r>>i;
    }

};

int main()
{
    complex a,b,c,d,e,f;

    a.input();
```

```
        b.input();

        a = a + b;
        a.display();

        c.input();
        d.input();

        c = c - d;
        c.display();

        e.input();
        f.input();

        e.display();
        f.display();
        if((e==f)!=0)
            cout<<"both are same";
        else
            cout<<"both are not same";

        return 0;
    }

    */

    /*
#include <iostream>
using namespace std;

class unary //15
{
    int m,n;

    public :

        unary()
        {
            m=8,n=9;
        }

        void operator-()
        {
            m=-m;
            n=-n;
        }
    }
```

```
void operator--()
{
    --m;
    --n;
}

void operator--(int)
{
    m--;
    n--;
}

void operator++()
{
    ++m;
    ++n;
}

void operator++(int)
{
    m++;
    n++;
}

void show()
{
    cout<<m<<" "<<n<<endl;
}

};

int main()
{
    unary a,b,c,d,e;

    a.show();
    -a;
    a.show();

    b.show();
    --b;
    b.show();

    c.show();
    ++c;
    c.show();

    d.show();
    d--;
    d.show();
}
```

```
        e.show();  
        e++;  
        e.show();  
  
        return 0;  
    }  
    */
```

```
/*  
#include <iostream>  
using namespace std;  
  
int main()  
{  
  
        return 0;  
}  
*/
```